

An Explainable Ensemble Model for Credit Card Fraud Detection Using Advanced Data Balancing and Feature Selection Technique

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Abstract—The rise of eCommerce and digital banking has transformed financial transactions, making online payments an essential part of modern economies. However, the increasing reliance on credit cards has led to a surge in fraudulent activities, highlighting the critical need for efficient fraud detection to secure digital payment systems. Traditional methods often struggle with accuracy, interpretability, and adaptability due to data imbalance and high-dimensional feature spaces. To address these challenges, this study introduces an advanced machine learning pipeline that integrates ensemble learning with optimized data balancing and feature selection techniques. Nine machine learning models were trained, and the top three—Random Forest, Gradient Boosting Machine (GBM), and CatBoost Classifier—were selected based on accuracy and combined into a Voting Classifier to leverage their complementary strengths. The ensemble model achieved 98.11% accuracy, 96.60% precision, 99.74% recall, and 98.14% F1-score, demonstrating its effectiveness in fraud detection. Additionally, Explainable Artificial Intelligence (XAI) techniques, LIME and SHAP, were employed to enhance model interpretability and identify key fraud-related features. This study contributes to improving fraud detection in highly imbalanced datasets, strengthening digital payment security, and increasing trust in online transactions.

Index Terms—Digital Payment Security, Credit Card Fraud Detection, Imbalanced Data Handling, Machine Learning, Explainable AI (XAI), Ensemble Learning.

I. INTRODUCTION

The rapid expansion of e-commerce and digital banking has made online payments a cornerstone of modern economies, with credit cards serving as a widely accepted medium for both online and offline transactions. However, this widespread adoption has also fueled a surge in fraudulent activities, creating significant challenges for financial institutions, businesses, and consumers [1]. Credit card fraud, which involves the unauthorized use of card information to obtain goods, services, or funds illegally, resulted in approximately \$32 billion in financial losses worldwide in 2024 alone [2]. Projections indicate that these losses could exceed \$43 billion by 2026, as online transactions continue to grow at an annual rate of 35%,

further increasing vulnerabilities to fraud and underscoring the urgent need for advanced fraud detection mechanisms [3].

Credit card fraud takes on several forms, each with its own unique challenges [1]. Card-not-present fraud occurs when criminals use stolen card details to perform unauthorized transactions online or over the phone without needing the actual card. In contrast, card-present fraud involves the physical theft or cloning of a card—often via skimming devices at point-of-sale terminals to execute fraudulent in-person purchases. Additionally, identity theft fraud happens when criminals obtain personal information to impersonate a victim, thereby opening new accounts or conducting unauthorized transactions on existing ones [4]. To mitigate these threats, traditional methods have been put in place, such as rule-based systems that flag transactions based on predetermined criteria, anomaly detection techniques that identify deviations from typical spending behaviors, statistical models that pinpoint unusual outliers in transaction data, and behavioral profiling that monitors spending patterns, locations, and transaction timings for fraud signs [4]. Traditional methods, rely on static rules and manual updates, fall short in adapting to new fraud methods and processing high volumes of real-time data, underscoring the need for dynamic ML-based approaches [3].

Machine learning provides powerful techniques for fraud detection by uncovering subtle patterns in large datasets. Traditional models like Logistic Regression, SVM, and Decision Trees offer robust baselines, while ensemble methods—such as bagging, boosting, and stacking—integrate multiple models to tackle imbalanced datasets and reduce false positives and negatives [4]. Advanced boosting algorithms like XGBoost, GBM, AdaBoost, LightGBM, and CatBoost further refine predictions by sequentially correcting errors, enabling real-time detection of emerging fraudulent patterns [5]. However, XAI techniques play a critical role in enhancing model interpretability and transparency [6]. LIME (Local Interpretable Model-agnostic Explanations) breaks down individual predictions by highlighting the key features that influence each decision [7]. On the

other hand, SHAP (SHapley Additive exPlanations) provides a broader, global view by assigning importance values to each feature based on their overall contribution to the model's predictions [6]. Together, these techniques not only improve the trustworthiness of fraud detection systems but also enhance their accuracy by enabling a clearer understanding of model behavior and facilitating more informed decision-making.

This paper presents an explainable ensemble model for credit card fraud detection, combining advanced data balancing, feature selection, and ensemble learning with LIME and SHAP. The approach enhances accuracy, mitigates detection challenges, and provides transparent insights for financial institutions. However, the rest of this paper is structured as follows. Section 2 presents the related work. Section 3 provides the experimental methodology, including results. Finally, conclusions and discussions are presented in Section 4.

II. RELATED WORK

The increasing prevalence of credit card fraud has driven extensive research into developing robust fraud detection models. Various studies have explored machine learning and deep learning techniques to enhance detection accuracy, mitigate data imbalance issues, and improve model interpretability.

Madhurya et al. [8] focused on handling imbalanced datasets in fraud detection using Support Vector Machine (SVM), Gradient Boost, and Random Forest. Through SMOTE-based balancing and GridSearchCV for hyperparameter tuning, their K-Nearest Neighbors (KNN) model achieved 95.84% accuracy, 96.61% sensitivity, and 74.46% specificity. Despite performance variability, their approach improved adaptability in fraud classification. Additionally, Yu et al. [9] introduced an advanced Transformer model for fraud detection, optimizing dataset balancing and training processes. The XGBoost achieved 95% accuracy. However, the dataset size and distribution required further validation for real-world applications.

Data imbalance remains a critical challenge in fraud detection, as fraudulent transactions are significantly fewer than legitimate ones. Ileberi and Sun [10] conducted a comparative study of twelve machine learning algorithms, employing SMOTE for data balancing and AdaBoost with majority voting for fraud detection. Their random forest model achieved 95% accuracy and 91% sensitivity, highlighting its effectiveness.

Recent advancements in deep learning have led to the development of more sophisticated fraud detection models. Kewei et al. [11] developed a hybrid deep learning model utilizing the IEEE-CIS fraud dataset (over 1 million transactions). Their model underwent rigorous feature engineering and data preprocessing, outperforming traditional machine learning models, including Naïve Bayes and SVM, with an accuracy of 95.8%. The study emphasized the role of memory compression and mixed precision in enhancing deep learning performance. Additionally, Najadat et al. [12] applied a BiLSTM-MaxPooling-BiGRU-MaxPooling approach to fraud detection, using the IEEE-CIS dataset from Kaggle. Their model outperformed six machine learning classifiers—Naïve Bayes, Voting, AdaBoost, Random Forest, Decision Tree, and Logistic Regression—achieving an accuracy of 91.37%.

Ensemble methods enhance machine learning performance by combining multiple models to improve accuracy, robustness, and generalization. Hussein et al. [13] proposed a

stacking ensemble technique combining Fuzzy Rough Nearest Neighbor (FRNN), Sequential Minimal Optimization (SMO), and Logistic Regression. Using UCI credit approval datasets, their model achieved an 84.90% detection rate and 86.20% specificity, though limitations included small dataset size and lack of direct comparison with state-of-the-art models.

However, while some recent studies have reported exceptional accuracy, even reaching up to 100% [9], [14], most have relied on the dataset with a limited set of parameters. This raises concerns about the potential exclusion of critical information, which could impact the real-world applicability and robustness of these models. Various machine learning techniques, including Support Vector Machine (SVM), k-Nearest Neighbor (KNN), Artificial Neural Networks (ANN), Logistic Regression, LightGBM, and Convolutional Neural Networks (CNNs), have been employed in these studies [15], [16]. For example, Bagga et al. [17] compared nine machine learning models and used ADASYN to address data imbalance. Their ensemble learning model achieved 99.99% accuracy, with pipelining reaching 99.99%. Despite these impressive results, the dataset remained imbalanced, raising concerns about its impact on classification performance and the model's ability to generalize effectively.

The reviewed studies demonstrate significant advancements in credit card fraud detection through machine learning methodologies. However, challenges such as dataset imbalance, feature selection limitations, model explainability, generalizability, computational costs, and adaptability to evolving fraud patterns persist. This study focused to address these issues by improving feature selection techniques and enhancing generalization capabilities to ensure fraud detection models remain effective in real-world applications.

III. METHODOLOGY

The proposed system follows a structured methodology for detecting fraudulent credit card transactions by integrating data preprocessing, feature selection, ensemble learning, and explainability techniques. As depicted in Figure 1, the workflow consists of four key stages: dataset preprocessing, feature selection, model training with an ensemble of classifiers, and model interpretability using LIME and SHAP [18]. Initially, raw transaction data undergoes preprocessing, which includes handling missing values, feature scaling, and addressing class imbalance through data augmentation with Generative Adversarial Networks (GANs). Principal Component Analysis (PCA) is then applied to reduce dimensionality while preserving critical information. The refined features are used to train multiple machine learning models, and the most effective classifiers are combined into an ensemble model to enhance fraud detection accuracy. The following subsections provide a detailed explanation of each stage.

A. Dataset Description

The dataset originates from the Université Libre de Bruxelles (ULB) and focuses on big data mining and fraud detection, covering transactions from January 1, 2018, to December 31, 2020. It includes transactions conducted by 1,000 customers across 800 merchants. For further information, the dataset is available on the ULB website. The dataset is also publicly available to access in Kaggle [19].

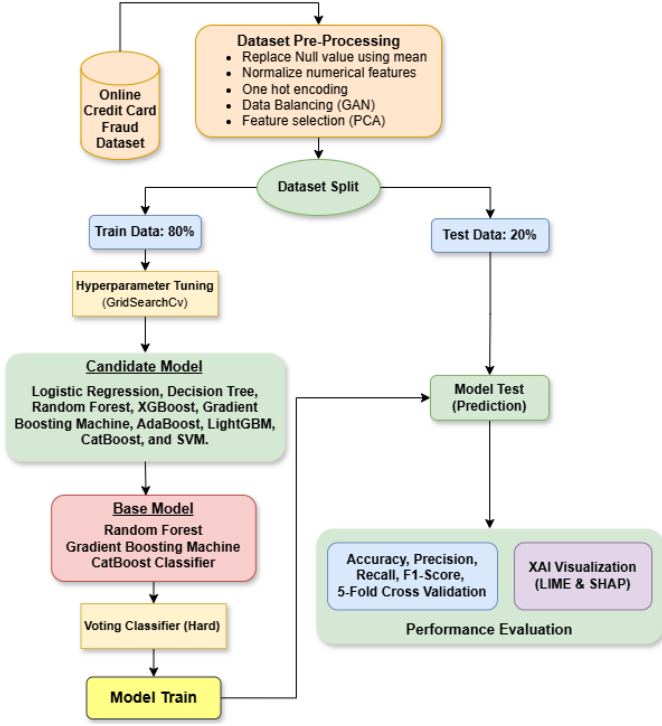


Fig. 1: Architecture of the Proposed Method.

The credit card fraud detection dataset consists of a total of 307,511 transactions, where 282,686 are legitimate, and 24,825 are fraudulent. Each transaction is represented by 122 numerical and categorical features, including transaction amount, timestamp, merchant ID, and various engineered features. The dataset is highly imbalanced, with fraudulent transactions comprising only 8.07% of the total samples. The objective is to accurately classify fraudulent transactions while minimizing false positives and ensuring the model generalizes well to unseen data.

B. Data Preprocessing

The dataset undergoes preprocessing to handle missing values, normalize numerical features, encode categorical variables, and address class imbalance. Missing values in numerical and categorical columns are imputed using the mean and mode, respectively. Numerical features are normalized using min-max scaling:

$$f_{\text{scaled}} = \frac{f - f_{\min}}{f_{\max} - f_{\min}} \quad (1)$$

where $f_{\min}$ and $f_{\max}$ denote the minimum and maximum values of feature f . Categorical features are one-hot encoded to ensure compatibility with machine learning models.

The dataset is highly imbalanced, with a significant disparity between fraud and non-fraud transactions. Due to the dataset's class imbalance, Generative Adversarial Networks (GANs) generate synthetic fraud transactions. GANs consist of a generator G , which maps a noise vector z from a Gaussian distribution to a synthetic feature vector $\hat{x}$,

$$\hat{x} = G(z; \theta_G) \quad (2)$$

where θ_G represents the generator's parameters. The discriminator D distinguishes real from synthetic fraud transactions, optimizing the loss function:

$$\min_G \max_D \mathbb{E}_{x \sim P_{\text{real}}} [\log D(x)] + \mathbb{E}_{z \sim P_z} [\log(1 - D(G(z)))] \quad (3)$$

where P_{real} is the distribution of real fraud transactions, and P_z is the noise distribution. GANs outperform traditional oversampling methods like SMOTE by capturing complex fraud patterns, improving model generalization.

With 122 features, the dataset poses computational challenges and risks of redundancy, Noise and overfitting. Principal Component Analysis (PCA) reduces dimensionality while preserving variance. Given the covariance matrix Σ , PCA solves:

$$\Sigma v = \lambda v \quad (4)$$

where v are principal component directions and λ are eigenvalues. Features are selected based on their variance contribution, reducing the dataset to 44 principal components, enhancing efficiency while retaining key fraud patterns for model training.

C. Ensemble Model Building

The classification process involves training multiple machine learning models to detect fraudulent transactions effectively. The dataset is split 80:20 into training and testing sets, ensuring models learn meaningful patterns while being evaluated on unseen data. To enhance robustness and prevent overfitting, five-fold cross-validation is applied, where models are trained on four subsets and validated on the fifth, averaging five times repeated results for a more generalized performance. The cross-validation loss is computed as the average error across these five subsets, ensuring that models do not overfit specific portions of the data.

Nine machine learning models are initially trained: Logistic Regression, Decision Tree, Random Forest, XGBoost, Gradient Boosting Machine (GBM), AdaBoost, LightGBM, CatBoost, and Support Vector Machine (SVM). Each model learns to differentiate between fraudulent and non-fraudulent transactions based on the transformed feature set obtained after preprocessing. Once trained, these models are evaluated based on their predictive performance using standard classification metrics, including accuracy, precision, recall, and F1-score. The evaluation ensures that the models generalize well to unseen data and do not overfit the training set.

The top three classifiers—Random Forest, GBM, and CatBoost—are selected for ensemble learning. Models are combined into a Voting Classifier, where each independently predicts fraud likelihood, and the final decision follows a majority voting strategy. The classifier's prediction is represented as y_{model_i} , the final fraud prediction y_{final} is computed as,

$$y_{\text{final}} = \frac{1}{3} \sum_{i=1}^3 y_{\text{model}_i} \quad (5)$$

where individual predictions are binarized based on a pre-defined threshold. The ensemble approach enhances classification reliability by leveraging the strengths of multiple models, reducing variance, and improving fraud detection accuracy. Performance is evaluated using accuracy, precision, recall, and

F1-score to ensure effective fraud identification with minimal false positives. The ensemble approach enhances classification performance by leveraging the strengths of multiple models, increasing generalization, and improving detection accuracy.

D. Explainability with LIME and SHAP

While the ensemble model achieves high accuracy, its decision-making process lacks transparency. To address this, LIME and SHAP are applied to explain fraud classifications, providing insights into key features influencing predictions.

LIME approximates the complex ensemble model locally using an interpretable surrogate model, such as Logistic Regression or decision trees. For a given transaction x , LIME generates perturbed samples x' in the neighborhood of x and observes the model's predictions. The importance of each feature is determined by fitting a locally weighted linear model $g(x)$ that approximates the decision boundary around x , formulated as:

$$\hat{f}(x) = \arg \min_{g \in G} \mathbb{E}_{x'} [L(f, g, x')] \quad (6)$$

where $f(x)$ is the original model, $g(x)$ is the interpretable approximation, and L is a loss function that measures the difference between the two. LIME highlights key features contributing to fraud classification, enabling human analysts to interpret model predictions.

SHAP provides a global and local interpretability framework based on cooperative game theory. It assigns Shapley values to each feature, quantifying its contribution to a specific prediction. Given a model f , the Shapley value for a feature i is computed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (7)$$

where F is the set of all features, and S represents a subset without feature i . SHAP ensures fair attribution of each feature's influence on the model's decision.

By integrating both methods, this study enhances fraud detection transparency, allowing financial institutions to refine detection rules and improve transaction monitoring, see Figure 3 and 4. The combination of GAN-based augmentation, ensemble learning, and explainability techniques ensures a robust, interpretable, and trustworthy fraud detection framework.

IV. RESULTS AND DISCUSSION

This section presents a detailed evaluation of the models used for credit card fraud detection, comparing their performance under two settings: (1) Without Data Augmentation and Feature Selection, and (2) With Data Augmentation and Feature Selection. The first setting provides a baseline, while the second showcases the effectiveness of our advanced data augmentation and feature selection techniques in improving fraud detection. The results are summarized in Table I, and additional insights are provided through the confusion matrix (Figure 1) and explainability analysis using LIME (Figure 3).

Table I highlights a significant improvement in performance when applying data augmentation and feature selection. Without these techniques, models struggle to achieve high recall, a crucial metric in fraud detection due to the highly imbalanced

nature of the dataset. However, after applying these techniques, all models exhibit notable gains in accuracy, precision, recall, and F1-score, demonstrating the importance of preprocessing strategies in fraud detection systems. Among the individual models, Random Forest, Gradient Boosting (GBM), and CatBoost performed well even without augmentation, achieving F1-scores above 85%. However, their performance improved further with augmentation, particularly in the recall, which increased to 99.63% for GBM and 99.45% for CatBoost, ensuring fewer false negatives. The ensemble model delivered the highest performance across all metrics, with an accuracy of 98.11%, precision of 96.60%, recall of 99.74%, and an F1-score of 98.14%, demonstrating the effectiveness of combining multiple models for fraud detection.

Figure 2 illustrates the confusion matrix for the best-performing ensemble model. The model correctly classified 56,389 negative samples and 54,553 positive samples, with minimal misclassification. The false positive (149) and false negative (1,984) rates were low, further indicating the effectiveness of the model in distinguishing between diseased and normal cases. The high recall (99.74%) of the ensemble model indicates its capability to detect diseased cases effectively, which is critical in medical diagnostics to minimize false negatives. The slight presence of false positives suggests that the model errs on the side of caution, a desirable trait in medical applications where misclassification of diseased cases as healthy could have severe consequences.

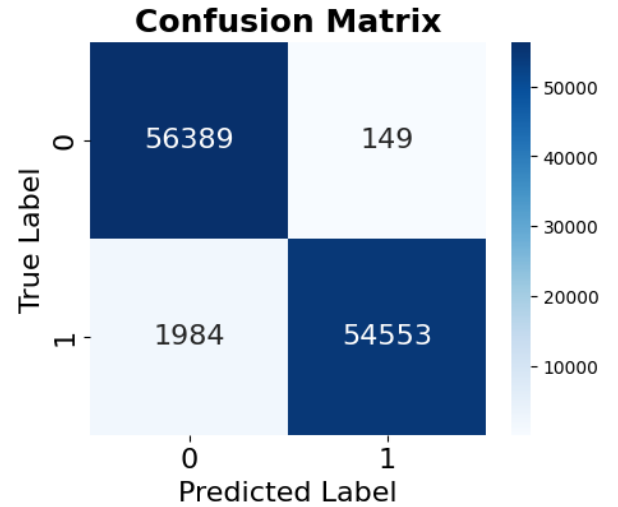


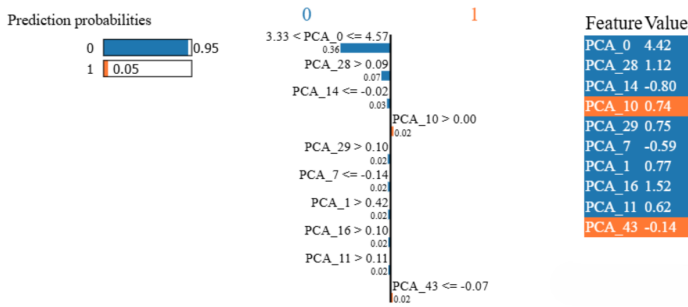
Fig. 2: Confusion Matrix of the Proposed Model's Prediction

To gain further insights into model decisions, LIME analysis was conducted to explain individual classifications. Figure 3 presents LIME explanations for both non-fraudulent (Class 0) and fraudulent (Class 1) transactions. The key contributing PCA features, such as PCA_0 , PCA_{28} , PCA_{33} , and PCA_9 , played a crucial role in determining the final prediction. In the case of non-fraudulent transactions (Figure 3a), PCA_0 had a strong positive contribution, whereas for fraudulent transactions (Figure 3b), PCA_0 exhibited a significantly negative impact, influencing the classification outcome.

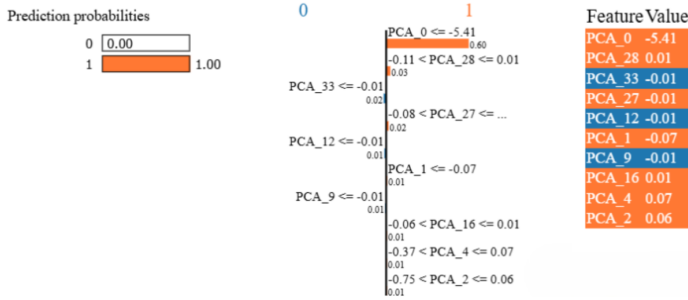
Figure 4 presents the SHAP summary plot, which highlights the most influential features contributing to the model's predictions. Feature 0 exhibits the highest impact on the model's

TABLE I: Performance Comparison of All Models with and without Data Augmentation and Feature Selection.

Model	Without Data Augmentation and Feature Selection				With Data Augmentation and Feature Selection			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
Logistic Regression	57.04%	56.70%	61.50%	59.00%	91.24%	89.92%	92.90%	91.39%
Decision Tree	80.73%	78.14%	85.34%	81.58%	91.36%	90.51%	92.41%	91.45%
Random Forest	85.37%	84.12%	87.20%	85.63%	95.68%	91.36%	99.98%	95.48%
XGBoost	84.98%	83.82%	86.68%	85.23%	94.96%	92.21%	98.21%	95.11%
GBM	85.54%	84.23%	87.46%	85.81%	96.11%	93.09%	99.63%	96.24%
AdaBoost	84.89%	83.69%	86.68%	85.16%	94.66%	91.74%	98.12%	94.82%
LightGBM	84.40%	83.08%	86.40%	84.71%	94.91%	91.89%	98.47%	95.07%
CatBoost	86.02%	84.66%	87.98%	86.29%	95.84%	92.75%	99.45%	95.98%
SVM	83.20%	87.42%	80.62%	83.88%	94.58%	91.74%	97.97%	94.75%
Proposed - Ensemble	87.19%	85.69%	89.28%	87.45%	98.11%	96.60%	99.74%	98.14%



(a) LIME Explanation for a Non-Fraudulent Prediction (Class 0)



(b) LIME Explanation for a Fraudulent Prediction (Class 1)

Fig. 3: LIME interpretability analysis. (a) Important features influencing a non-fraud prediction. (b) Important features contributing to a fraud prediction.

output for both Class 0 and Class 1, indicating its crucial role in classification decisions. The significant contribution of Feature 0 suggests that the model heavily relies on this feature when distinguishing between fraudulent and non-fraudulent transactions. Furthermore, several other features, such as Feature 2, Feature 28, and Feature 1, also contribute, albeit to a lesser extent. Notably, the color representation in the SHAP plot differentiates the classes, showing that the influence of certain features varies depending on the prediction category. Features like Feature 2 and Feature 28 show a small but consistent contribution across both classes, reinforcing their importance in decision-making.

This interpretability analysis confirms that the model makes informed decisions based on relevant features rather than random noise, ensuring reliability in clinical settings. Moreover, the use of dimensionality-reduced features (PCA components) suggests that the feature selection process effectively captured

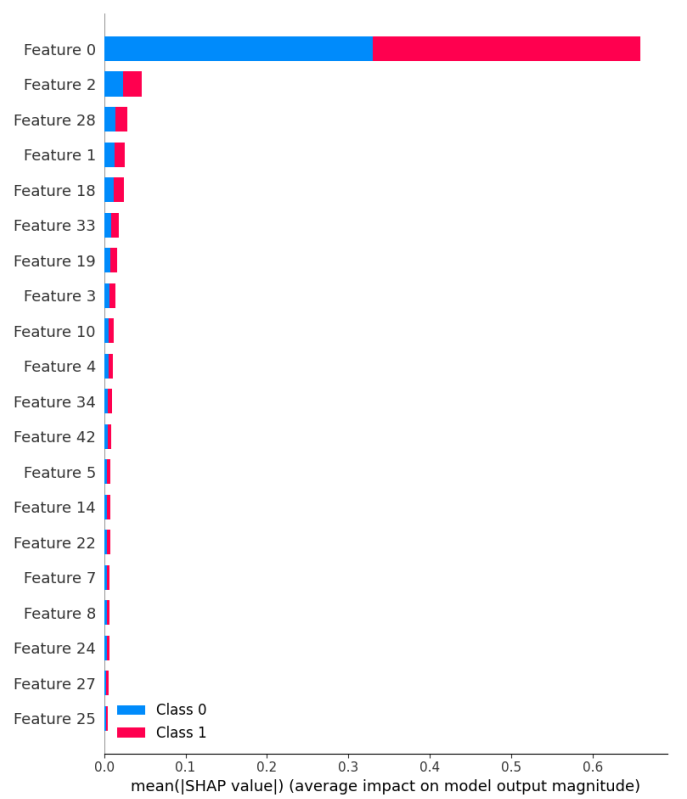


Fig. 4: SHAP Analysis of Feature Contributions in the Proposed Model's Predictions.

the most informative aspects of the dataset.

Table II presents a comparison of our proposed ensemble model with the state-of-the-art (SOTA) existing methods for credit card fraud detection having superior results. The comparison includes approaches proposed by Vihurskyi et al. [14], Yu et al. [9], Taha et al. [20], Alharbi et al. [16] and Kewei et al. [11], which employ machine learning models such as Random Forest, XGBoost, LightGBM, CatBoost, and SVM. We implemented all these existing methods on our dataset in the same computational environment and setup to ensure a fair comparison. The models are evaluated based on accuracy, precision, recall, and F1-score. Our proposed ensemble model achieves the highest accuracy of 98.11%, precision of 96.60%, and F1-score of 98.14%, while maintaining a significantly high recall of 99.74%. The high recall indicates that the model

effectively detects fraudulent transactions, which is essential for minimizing false negatives.

Compared to the best-performing baseline, CatBoost (Alharbi et al. [16]), our ensemble method improves accuracy and F1-score while maintaining a high recall, demonstrating the effectiveness of our advanced data augmentation and feature selection techniques. These results confirm that our approach enhances fraud detection performance by reducing misclassification errors and improving model robustness.

TABLE II: Comparison of Existing Methods and the Proposed Model for Credit Card Fraud Detection.

Ref	Model	Accuracy	Precision	Recall	F1-score
[20]	LightGBM	94.91%	91.89%	98.47%	95.07%
[14]	Random Forest	95.68%	91.36%	99.98%	95.48%
[9]	XGBoost	94.96%	92.21%	98.21%	95.11%
[16]	CatBoost	95.84%	92.75%	99.45%	95.98%
[11]	SVM	94.58%	91.74%	97.97%	94.75%
Proposed	Ensemble	98.11%	96.60%	99.74%	98.14%

V. CONCLUSION

This study presents an advanced credit card fraud detection approach that leverages ensemble learning, feature selection, and data augmentation to improve classification performance. The proposed model achieves 98.11% accuracy, outperforming state-of-the-art (SOTA) methods, with a high recall of 99.74%, ensuring effective fraud detection while minimizing false negatives. The interpretability analysis using LIME and SHAP highlights the key features influencing fraud predictions, enhancing model transparency and trustworthiness. Despite its effectiveness, the model has limitations. PCA-transformed features may reduce interpretability compared to raw transaction attributes, and synthetic data from GANs might not fully capture real-world fraud patterns. Additionally, the computational complexity of ensemble learning poses challenges for real-time deployment in high-volume transaction environments. Future work will explore hybrid deep learning techniques to further enhance fraud detection accuracy. Graph-based anomaly detection could improve the identification of complex fraud patterns, while lightweight architectures would be investigated for real-time feasibility. Additionally, adaptive learning mechanisms can be integrated to ensure the model remains robust against evolving fraud strategies.

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